RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

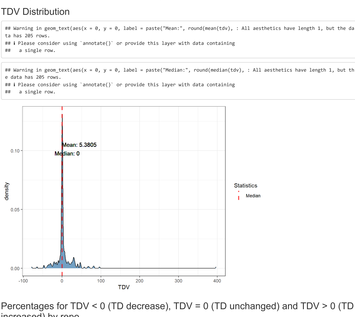
```
## Rows: 19,586
## Columns: 11
## $ repo              <chr> "helix", "helix", "helix", "helix", "helix", "heli…
## $ pr_number         <int> 1438, 1438, 1438, 1438, 1441, 1441, 1441, 1441, 14…
## $ rule              <chr> "java:S115", "java:S1068", "java:S1068", "java:S137…
## $ severity          <chr> "MINOR", "MINOR", "MINOR", "CRITICAL", "INFO", "MI…
## $ file              <chr> "helix-core/src/main/java/org/apache/helix/control…
## $ type              <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status            <chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O…
## $ debt              <int> 20, 2, 2, 11, 0, 2, 2, 1, 2, 1, 2, 2, 2, 5, 2, …
## $ n_nloc_affected_file <int> 102, 102, 102, 102, 127, 127, 127, 127, 127, 127, …
## $ complexity        <int> 20, 20, 20, 20, 19, 19, 19, 19, 19, 19, 19, 19, 19…
## $ origin            <chr> "NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTI…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 205 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 205 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

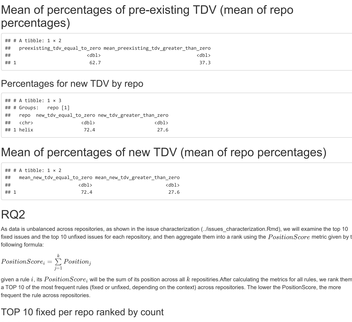


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 205 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 205 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 × 4
##   repo  tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>               <dbl>              <dbl>                 <dbl>
## 1 helix                  20               44.9                  35.1
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                     <dbl>
## 1                       20                   44.9                      35.1
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 × 3
##   repo  preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                         <dbl>                             <dbl>
## 1 helix                          62.7                              37.3
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          62.7                                   37.3
```

Percentages for new TDV by repo

```
## # A tibble: 1 × 3
## # Groups:   repo [1]
##   repo  new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                 <dbl>                     <dbl>
## 1 helix                  72.4                      27.6
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                       72.4                           27.6
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositories.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##    rank helix
##   <int> <chr>
##  1     1 java:S1874  - 102
##  2     2 java:S116   -  97
##  3     3 java:S2293  -  63
##  4     4 java:S1135  -  36
##  5     5 java:S3776  -  33
##  6     6 java:S3740  -  31
##  7     7 java:S1186  -  24
##  8     8 java:S1172  -  22
##  9     9 java:S1192  -  20
## 10    10 java:S2864  -  15
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule      position_score
##    <chr>              <dbl>
##  1 java:S1874           122
##  2 java:S116            123
##  3 java:S2293           124
##  4 java:S1135           125
##  5 java:S3776           126
##  6 java:S3740           127
##  7 java:S1186           128
##  8 java:S1172           129
##  9 java:S1192           130
## 10 java:S186            131
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##    rank helix
##   <int> <chr>
##  1     1 java:S116   - 4208
##  2     2 java:S1874  - 2026
##  3     3 java:S1135  - 1015
##  4     4 java:S2293  - 1513
##  5     5 java:S115   - 1058
##  6     6 java:S3776  -  684
##  7     7 java:S1192  -  640
##  8     8 java:S1133  -  613
##  9     9 java:S1123  -  498
## 10    10 java:S2864  -  486
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule      position_score
##    <chr>              <dbl>
##  1 java:S116            122
##  2 java:S1874           123
##  3 java:S1135           124
##  4 java:S2293           125
##  5 java:S115            126
##  6 java:S3776           127
##  7 java:S1192           128
##  8 java:S1133           129
##  9 java:S1123           130
## 10 java:S2864           131
```

Outliers

Higher TDD

```
## # A tibble: 205 × 5
##    repo  pr_number total_debt total_ncloc   tdd
##    <chr>     <int>      <int>       <int> <dbl>
##  1 helix      1975        261          33  7.91
##  2 helix      2488        126          51  2.47
##  3 helix      2421          5           3  1.67
##  4 helix      1531        433         298  1.49
##  5 helix      2249        526         367  1.43
##  6 helix      1620        797         586  1.36
##  7 helix      2223         60          46  1.30
##  8 helix      2618       3056        2375  1.29
##  9 helix      1753        109          86  1.27
## 10 helix      1678        159         136  1.17
## # ℹ 195 more rows
```

Higher TDV

```
## # A tibble: 205 × 3
##    pr_number repo    tdv
##        <int> <chr> <int>
##  1      2249 helix   397
##  2      1975 helix    96
##  3      2546 helix    82
##  4      1753 helix    80
##  5      2488 helix    69
##  6      2687 helix    66
##  7      1732 helix    57
##  8      1550 helix    47
##  9      1685 helix    47
## 10      2736 helix    47
## # ℹ 195 more rows
```

Lower TDV

```
## # A tibble: 205 × 3
##    pr_number repo    tdv
##        <int> <chr> <int>
##  1      2021 helix   -79
##  2      2149 helix   -65
##  3      1693 helix   -47
##  4      2180 helix   -47
##  5      1850 helix   -39
##  6      1487 helix   -37
##  7      2470 helix   -35
##  8      2500 helix   -34
##  9      2456 helix   -32
## 10      2585 helix   -30
## # ℹ 195 more rows
```

Extra

Number of PRs with pre-existing TD

```
##    n
## 1 201
```

Number of java:S1192 issues that affect test code

```
##  n
## 1 0
```

Number of java:S1192 issues that affect not test code

```
##    n
## 1 660
```

Summary NCLOC

```
##      ncloc
##  Min.   :   3
##  1st Qu.:  294
##  Median :  713
##  Mean   : 1346
##  3rd Qu.: 1625
##  Max.   :15590
```